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RESEARCH ARTICLE

An Adaptive Modality Fusion Network for Stock Trend Prediction

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ABSTRACT Financial markets are increasingly shaped by structured indicators and unstructured news flows, making trend prediction a multimodal challenge under uncertainty. This paper introduces the Adaptive Modality Fusion Network (AMFNet), a deep learning architecture that integrates technical indicators with document-level sentiment extracted from multiple language models through a structured sentiment matrix, a Transformer encoder, and an uncertainty-aware fusion mechanism. We evaluate AMFNet on a ten-year dataset covering twelve U.S. large-cap equities and three emerging-market American Depositary Receipts (ADRs). Results show that AMFNet surpasses recent multimodal baselines in ternary trend classification, achieving an F1-score of 86.8% on 2023–2024 out-of-sample tests. Portfolio backtests, with and without transaction costs, confirm that predictive accuracy translates into profitability and risk-adjusted returns. Stress tests during the COVID-19 crash and the 2022 inflationary crisis demonstrate resilience under extreme shocks, while ADR results indicate cross-market portability. Attention-based weights provide interpretability by revealing dynamic shifts between sentiment and technical indicators. Altogether, AMFNet delivers accuracy, robustness, and transparency, offering a practical framework for forecasting and risk management.

INDEX TERMS Multimodal fusion, sentiment analysis, stock trend prediction, technical indicators, transformer networks.

I. INTRODUCTION

Forecasting stock market movements remains a core problem in financial modeling, complicated by market non-stationarity, behavioral noise, and the influence of diverse information streams, including price trends, macroeconomic signals, and investor sentiment [1], [2]. Despite growing interest in machine learning for financial prediction, the heterogeneity and volatility of market signals continue to pose challenges to conventional models.

Recent advances in deep learning, including Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNNs), and Transformers, have improved the modeling of structured time series [3], [4]. However, unstructured textual information -such as financial news- remains underexploited, even though it reflects real-time sentiment and expectations. Existing approaches often aggregate textual

signals at the daily level or treat them as static features, discarding variability between individual documents and overlooking uncertainty in sentiment interpretation [21].

To address these gaps, we propose the Adaptive Modality Fusion Network (AMFNet), a multimodal architecture that integrates technical indicators with fine-grained sentiment representations derived from multiple language models. AMFNet introduces three core innovations: (1) a structured sentiment matrix preserving document- and model-level diversity; (2) a Transformer-based encoder capturing inter-document semantic dependencies; and (3) an uncertainty-aware attention mechanism that adaptively fuses sentiment and technical signals based on contextual relevance.

Our main contributions are as follows:

- A document-level sentiment representation that preserves diversity across documents and models, enabling rich, non-aggregated textual encoding.

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- An uncertainty-aware fusion module that modulates modality importance using inter-model disagreement as a proxy for confidence, enhancing robustness under noisy or conflicting news flows.
- Comprehensive empirical evidence that AMFNet outperforms recent multimodal baselines across twelve U.S. large-cap equities and three emerging-market ADRs, covering diverse sectors and geographies over a ten-year horizon.
- Portfolio-based evaluation, reported both with and without transaction costs, demonstrating that statistical gains translate into competitive profitability and risk-adjusted returns.
- Interpretable attention outputs that reveal dynamic reliance on sentiment versus technical features across sectors and market regimes, supporting transparency and informed financial decision-making.

This study contributes a flexible and interpretable framework for multimodal financial forecasting, combining structured technical analysis with fine-grained sentiment modeling. AMFNet not only achieves superior predictive accuracy but also demonstrates resilience during crises such as the COVID-19 crash and the 2022 inflationary shock, while generalizing effectively to emerging-market equities. Its design is particularly suitable for integration into decision support systems, where reliable short-term signals and modality-level interpretability are essential for portfolio rebalancing, trading strategy adjustment, or institutional risk monitoring.

The rest of the paper is organized as follows. Section II surveys related literature. Section III details the proposed model. Section IV describes data and training procedures. Section V presents results and interpretability insights. Section VI concludes with future directions.

II. RELATED WORK

Stock trend forecasting has evolved significantly, progressing from statistical methods to advanced deep learning and multimodal fusion architectures.

Early approaches relied on linear models such as ARIMA and GARCH [5], [6], which were effective for stationary series but were unable to capture the inherent non-linearity and volatility of financial markets. To overcome these limitations, classical machine learning models such as SVMs, decision trees, and random forests were introduced [7], [8]. While they allowed more flexibility in handling tabular features, they could not model temporal dynamics and remained heavily dependent on feature engineering.

The emergence of deep learning models, such as LSTM and GRU, has offered improvements in the modeling of sequential dependencies [9], [10]. However, their performance degraded over longer sequences because of gradient vanishing and their sequential processing nature. Attention-based enhancements [11] and dual-stream architectures partially mitigated these drawbacks. More recently, convo-

lutional temporal networks have been adapted for finance: Biswas et al. [36] designed a dual-output TCN with attention to jointly predict prices and risk measures, highlighting the value of attention for risk-aware forecasting, though their method remains unimodal.

Transformers fundamentally changed time-series modeling by enabling parallelism and capturing long-term dependencies via self-attention [12]. StockFormer [13], for instance, integrated FinBERT sentiment with technical indicators, outperforming LSTM by 18.63% in directional accuracy. Beyond stock markets, Tuncer et al. [35] applied Vision Transformers to 2D encodings of technical indicators and demonstrated superior Sharpe ratios across ETFs in realistic backtests with transaction costs, showing the suitability of attention-based architectures for financial forecasting. Nevertheless, their framework is restricted to purely technical signals. In parallel, graph-based temporal learning has gained traction. Uygun and Sefer [41] benchmarked spatio-temporal graph neural networks (ST-GNNs) on Forex and cryptocurrencies, reporting consistent gains over LSTM and VAR. Amiri et al. [37] further combined dynamic time warping-based GCNs with LSTM-attention for the energy sector, surpassing classical baselines and highlighting the benefit of capturing inter-stock relationships. These works confirm the potential of graph architectures, but they remain unimodal and overlook sentiment uncertainty.

To address such limitations, multimodal fusion models have emerged. MSGCA [14] combines MLPs, large language models (LLMs), and graph attention networks (GATs) through gated cross-attention, achieving up to 31.6% improvement in MCC. Similarly, MSMF [15] introduces multi-scale encoders and modality completion mechanisms to handle heterogeneous inputs, demonstrating strong results in both return and movement prediction. The Higher Order Transformer [16] employs Kronecker-decomposed attention over multimodal time series, but its reliance on dated datasets and limited text sources (e.g., Twitter) hinders practical deployment. Zhen et al. [34] further validated the incremental value of sentiment by combining multi-feature selection with aggregated news and social sentiment for stock price regression, though their method lacks document-level representation and uncertainty modeling.

Hybrid methods that integrate sentiment and technical indicators have also proven effective. Reference [18] combined LSTM with ARIMA and TextBlob-derived sentiment, reaching 85% accuracy. Reference [21] proposed a deep fusion model that integrates historical prices, technical indicators, and news sentiment using LSTM and CNN layers, achieving competitive accuracy and robustness across market regimes. Similarly, [19] leveraged FinBERT and ChatGPT-4o sentiment fusion with technical features, achieving $R^2 = 0.9644$ and simulated returns of 42.11%. Extensions to cryptocurrencies have also been explored: Celik and Sefer [39] integrated on-chain transaction motifs with GNN-LSTM architectures, significantly improving regression and directional classification for BTC/ETH.

Belcastro et al. [38] combined market and large-scale social-media data, yielding highly profitable trading strategies (+117% net of costs), underscoring the relevance of social signals despite their noise. In contrast, Islam et al. [40] benchmarked traditional classifiers (logistic regression, RF, XGBoost) on BTC/ETH and achieved only modest accuracies (56%), illustrating the difficulty of crypto forecasting and motivating uncertainty-aware multimodal fusion.

Another hybrid paradigm leverages graph structures beyond sentiment. Liu [17] proposed a multimodal factor model combining textual, numerical, and social indicators using LSTM, BERT, and GNN, achieving $R^2 = 0.82$. Their approach improves robustness via attention-based fusion but does not address real-time constraints or interpretability.

Finally, recent reviews highlight the potential of LLMs for stock selection and financial modeling. Reference [20] benchmarked LLaMA2-based models and found them competitive with traditional ML baselines. However, such models often omit modality interaction mechanisms and are not yet optimized for uncertainty-aware forecasting.

Despite these advances, the existing literature reveals several persistent challenges:

- **Static fusion strategies:** Most multimodal systems use fixed fusion rules, failing to adapt to context or market regime;
- **Lack of uncertainty modeling:** Sentiment disagreement across sources is rarely quantified or leveraged;
- **Limited interpretability:** Few models expose modality-wise importance or enable sample-level diagnostics;
- **Sector- and asset-insensitive modeling:** Modality importance often varies across sectors or asset classes (e.g., equities vs. crypto), yet this is rarely accounted for.

To bridge these gaps, we propose AMFNet, an Adaptive Modality Fusion Network that dynamically fuses modalities based on volatility, sentiment uncertainty, and sector-specific

signals. Unlike prior works, AMFNet provides interpretability via per-modality attention scores.

III. PROPOSED METHODOLOGY

The Adaptive Modality Fusion Network (AMFNet) is a modular architecture for daily stock trend prediction that integrates structured technical indicators and document-level sentiment signals. It combines a Transformer-based encoder to model dependencies among news documents published on the same day, a divergence-based uncertainty feature to reflect sentiment disagreement across models, and a context-aware attention mechanism for adaptive modality fusion.

An overview of the framework is provided in Fig. 1, and each component is described in the following subsections.

A. PROBLEM FORMULATION

Let $\{(\mathbf{x}_t^{\text{tech}}, \mathbf{S}_t)\}_{t=1}^T$ represent a multimodal time series, where $\mathbf{x}_t^{\text{tech}} \in \mathbb{R}^d$ is a vector of d technical indicators, and $\mathbf{S}_t \in \mathbb{R}^{n \times m}$ denotes the sentiment matrix on day t , where n is the number of financial news items and m the number of sentiment models applied.

The objective is to predict the direction of the asset's return on the next trading day ($t + 1$), based on the information available at the time t . The return is computed as follows:

$$r_{t+1} = \frac{P_{t+1} - P_t}{P_t}, \quad (1)$$

where P_t denotes the closing price at time t . The corresponding label $y_t \in \{-1, 0, +1\}$ is assigned using a threshold $\delta > 0$ to eliminate insignificant fluctuations:

$$y_t = \begin{cases} +1 & \text{if } r_{t+1} > \delta \\ -1 & \text{if } r_{t+1} < -\delta \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

This thresholding mechanism allows the model to focus on meaningful directional shifts while filtering out market noise. The value of δ can be adjusted based on the volatility of the asset or the investor's risk preferences.

The learning objective is to approximate a function:

$$f_\theta : \mathbb{R}^d \times \mathbb{R}^{n \times m} \rightarrow \{-1, 0, +1\}, \quad \hat{y}_t = f_\theta(\mathbf{x}_t^{\text{tech}}, \mathbf{S}_t), \quad (3)$$

where θ denotes the learnable parameters of the model.

Training is performed by minimizing the categorical cross-entropy loss over N examples:

$$\min_{\theta} \mathcal{L}(\theta) = -\frac{1}{N} \sum_{t=1}^N \sum_{c \in \{-1, 0, +1\}} \mathbb{1}_{[y_t=c]} \log p_\theta(y_t = c \mid \mathbf{x}_t^{\text{tech}}, \mathbf{S}_t) \quad (4)$$

where $p_\theta(y_t = c \mid \cdot)$ is the predicted probability for class c at time t .

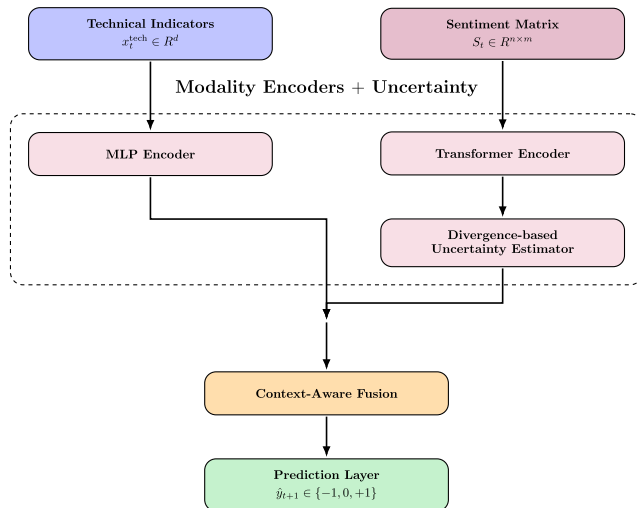


FIGURE 1. Overview of the AMFNet-based predictive framework.

B. MODEL ARCHITECTURE

The Adaptive Modality Fusion Network is a dual-branch architecture designed to encode heterogeneous financial signals for the joint prediction of the next day's stock trend. The model is built around three core properties: robustness to market regime changes via adaptive attention, interpretability through explicit modality weighting, and computational efficiency suitable for daily-scale inference.

At each time step t , the model receives two synchronized inputs:

- A vector of technical indicators $\mathbf{x}_t^{\text{tech}} \in \mathbb{R}^d$,
- A sentiment matrix $\mathbf{S}_t \in \mathbb{R}^{n \times m}$, where n is the number of financial news items on day t , and m denotes the number of sentiment models used.

These inputs are independently processed through dedicated encoders, and their latent representations are later fused through a context-aware attention mechanism to produce the final prediction.

1) MODALITY-SPECIFIC ENCODERS

AMFNet processes each modality through a dedicated encoder optimized for its respective data structure and learning dynamics.

The vector of technical indicators $\mathbf{x}_t^{\text{tech}} \in \mathbb{R}^d$ is projected into a latent representation using a two-layer multilayer perceptron (MLP):

$$H_{\text{tech}} = \phi_{\text{tech}}(\mathbf{x}_t^{\text{tech}}). \quad (5)$$

This approach is motivated by previous findings in financial modeling, where MLPs have been shown to effectively capture nonlinear interactions among low-dimensional numeric features, while maintaining low computational complexity [22].

For the sentiment modality, the matrix $\mathbf{S}_t \in \mathbb{R}^{n \times m}$ captures the outputs of m sentiment models applied to n financial news items from day t . Each row represents the sentiment profile of a document, preserving information at the document level rather than aggregating signals across the day. This retains fine-grained sentiment cues and inter-document variation, which are especially important during volatile market conditions.

The entire matrix is treated as a non-temporal sequence input to a Transformer encoder:

$$H_{\text{sent}} = \text{Transformer}(\mathbf{S}_t). \quad (6)$$

This setup enables the model to learn latent dependencies across documents, such as semantic alignment or contrast, which are often informative despite the absence of temporal ordering. The design is inspired by hierarchical attention networks [23] and recent attention-based architectures applied to financial text processing, such as the multi-view FinBERT-based HAN model proposed in [24].

We apply mean pooling over the Transformer outputs to obtain a fixed-length sentiment representation, as it yields

stable and reliable embeddings in variable-length sequences, particularly when labeled data are limited [25].

In addition to the main embeddings, AMFNet integrates an uncertainty signal u_t to capture the divergence between sentiment models. This signal is defined as the average standard deviation of normalized sentiment scores per document:

$$u_t = \frac{1}{n} \sum_{i=1}^n \text{std}(\mathbf{S}_t[i, :]). \quad (7)$$

Before computing this metric, each score is min-max normalized to the $[-1, +1]$ range to ensure comparability between models. The resulting scalar provides a proxy for epistemic uncertainty, particularly useful when sentiment models disagree due to conflicting signals or model biases.

This scalar score is concatenated to the sentiment embedding to form an augmented representation:

$$H_{\text{sent}}^* = [H_{\text{sent}} \parallel u_t], \quad (8)$$

where $\parallel$ denotes vector concatenation. This enriched representation allows the model to modulate the contribution of sentiment information based on its estimated reliability.

Since u_t is a scalar and $H_{\text{sent}} \in \mathbb{R}^d$, the concatenated vector $[H_{\text{sent}} \parallel u_t]$ lies in $\mathbb{R}^{d+1}$. To ensure compatibility with H_{tech} during fusion, we apply a linear projection:

$$H_{\text{sent}}^* = W_s [H_{\text{sent}} \parallel u_t] + b_s, \quad H_{\text{sent}}^* \in \mathbb{R}^d, \quad (9)$$

where $W_s \in \mathbb{R}^{d \times (d+1)}$ and $b_s \in \mathbb{R}^d$ are learnable parameters.

2) CONTEXT-AWARE FUSION

To jointly leverage both modalities in a flexible and interpretable manner, AMFNet employs an attention-based fusion layer that dynamically adjusts the contribution of each input. This is particularly important in financial contexts, where the relative informativeness of technical indicators and textual sentiment can change depending on market conditions.

The fusion mechanism receives the technical embedding H_{tech} and the sentiment representation H_{sent}^* , which includes the uncertainty signal as defined in the encoder module.

An attention gate learns to weight the two sources based on their joint representation:

$$\alpha = \text{softmax}(W_{\text{att}}[H_{\text{tech}} \parallel H_{\text{sent}}^*] + b_{\text{att}}), \quad (10)$$

$$H_{\text{fused}} = \alpha_1 H_{\text{tech}} + \alpha_2 H_{\text{sent}}^*, \quad (11)$$

where W_{att} and b_{att} are trainable parameters, and the attention weights $\alpha = (\alpha_1, \alpha_2)$ satisfy $\alpha_1 + \alpha_2 = 1$.

Unlike early or late fusion through simple concatenation, this formulation allows the model to learn context-dependent preferences between modalities, improving robustness and interpretability. Similar attention-based fusion mechanisms have been shown to outperform naive integration strategies in multimodal learning tasks [26].

3) PREDICTION LAYER

The final fused vector H_{fused} is passed through a classification head to generate the predicted probabilities on the three target classes: Buy (+1), Neutral (0), and Sell (−1). This is implemented as a fully connected layer followed by a softmax activation:

$$\hat{y}_t = \text{softmax}(\psi(H_{\text{fused}})), \quad (12)$$

where $\psi(\cdot)$ denotes a linear transformation that projects the fused representation into the output space $\mathbb{R}^3$.

This design enables end-to-end optimization via cross-entropy loss, aligning with the ternary classification objective defined in the problem formulation. The use of softmax ensures probabilistic interpretability and compatibility with standard evaluation metrics such as accuracy and F1-score.

C. INTERPRETABILITY AND PRACTICAL PROPERTIES

AMFNet is designed to balance predictive performance with interpretability and computational tractability, key requirements in financial applications. Its fully feedforward architecture enables efficient inference without recurrence or sequential decoding, making it well-suited for real-time deployment in daily prediction pipelines.

Interpretability is achieved through the attention-based fusion mechanism, where the learned weights (α_1, α_2) explicitly indicate the model's reliance on each modality. This allows practitioners to analyze whether decisions are primarily driven by technical indicators or sentiment signals, promoting transparency and trustworthiness, particularly important in high-stakes financial environments.

All latent representations are embedded in a shared vector space of dimension $d = 64$, a choice empirically validated as a trade-off between expressiveness and efficiency. This dimensionality ensures stable training while keeping memory and computation costs low, consistent with guidelines for scalable deep financial models.

IV. EXPERIMENTAL SETUP

This section outlines the data sources, preprocessing pipeline, feature construction, training configuration, and evaluation protocol used to assess AMFNet's performance. All experiments were conducted in a controlled and reproducible environment, as summarized in Table 1.

TABLE 1. Computational environment.

Component	Description/Values
Machine configuration	Google Colab with NVIDIA Tesla T4 GPU
Environment	Python 3.10.6, PyTorch 2.0.1, Transformers

In all experiments, each configuration is repeated with five different random seeds, and we report the mean and standard deviation of test metrics across these runs using identical data splits and hyperparameter settings. To ensure that AMFNet remains suitable for real-time deployment despite its three-model sentiment module, the implementation leverages standard efficiency practices such as batched

inference and result caching, details of which are provided in the supplementary material.

A. DATA COLLECTION

1) DATA DESCRIPTION

We construct a multimodal dataset spanning January 2015 to December 2024, covering diverse market regimes including crises (e.g., COVID-19) and periods of recovery. This long-term horizon ensures exposure to varying volatility and economic conditions.

Twelve U.S. companies were selected across six major sectors—Technology, Finance, Healthcare, Energy, Industrials, and Consumer Staples—based on market capitalization, trading volume stability, and sectoral representativeness. We chose exactly two companies per sector to provide sufficient intra-sector variation while maintaining a balanced cross-sector composition.

Fewer companies per sector would risk underrepresenting sector-specific news dynamics. In contrast, a larger number would increase the volume and heterogeneity of textual data, raising computational cost and potentially diluting the evaluation focus. These criteria ensure liquidity, consistent news coverage, and heterogeneity in financial behavior, thus improving the generalizability of the model.

TABLE 2. Selected U.S. assets grouped by sector.

Sector	Company	Ticker
Technology	Apple Inc.	AAPL
	NVIDIA Corporation	NVDA
Finance	JPMorgan Chase & Co.	JPM
	Bank of America Corp.	BAC
Healthcare	Johnson & Johnson	JNJ
	Pfizer Inc.	PFE
Energy	Exxon Mobil Corp.	XOM
	Chevron Corp.	CVX
Industrials	Boeing Co.	BA
	Caterpillar Inc.	CAT
Consumer Staples	The Coca-Cola Company	KO
	Procter & Gamble Co.	PG

TABLE 3. Emerging-market ADRs listed on NYSE.

Company	Ticker
Infosys Ltd (India)	INFY
ICICI Bank Ltd (India)	IBN
Vale S.A. (Brazil)	VALE

Beyond this core set of U.S. large-cap equities, we expand the dataset geographically to include emerging markets, allowing us to examine cross-market robustness. Specifically, we include three American Depositary Receipts (ADRs) listed on the New York Stock Exchange: Infosys Ltd (INFY, India), ICICI Bank Ltd (IBN, India), and Vale S.A. (VALE, Brazil). For these assets, the evaluation period spans January 2020 to December 2024, ensuring both sufficient data availability and consistent news coverage. The firms were selected to satisfy three requirements: (i) high liquidity and trading volume stability, ensuring reliable technical

indicators, (ii) consistent coverage in English-language financial media, providing sufficient textual signals, and (iii) diversity across both sectors and geographies. Their NYSE listings enable direct retrieval of OHLCV data via the Yahoo Finance API, ensuring that the preprocessing and modeling pipeline remains identical to that used for U.S. assets. This controlled extension provides a natural test of AMFNet's portability across heterogeneous market environments without methodological changes.

In addition to sectoral and geographical diversity, the dataset also spans heterogeneous market regimes to ensure that AMFNet is tested under stable and turbulent conditions. In particular, it includes the COVID-19 crash (February–April 2020), when global equities experienced one of the fastest drawdowns in history due to pandemic-induced uncertainty [31]. It also covers the inflationary and energy crisis (June–December 2022), during which tightening of monetary policy and geopolitical shocks triggered persistent volatility and equity market corrections [32]. Incorporating these episodes is essential, as they provide a natural stress test for predictive robustness and allow us to evaluate whether the model can adapt to extreme shocks as well as to subsequent recovery phases.

2) STOCK MARKET DATA

We collected daily OHLCV data (Open, High, Low, Close, Volume) for all selected companies via the Yahoo Finance API. From these, a set of domain-standard technical indicators was computed to capture price trends, momentum, volatility, and volume-based behaviors. The indicators include: RSI, MACD, OBV, ATR, Bollinger Band Width, and ROC. All features are computed at time t using information available up to that day, ensuring no lookahead bias. The resulting vector constitutes the technical input $\mathbf{x}_t^{\text{tech}}$ to AMFNet.

3) FINANCIAL NEWS DATA

For each asset and trading day t , we collect financial news from institutional sources (e.g., Bloomberg, CNBC) and community-driven platforms (e.g., Reddit), considering only articles published during market hours for temporal alignment. Both headline and full body are used to capture richer semantic signals.

Each document is scored by three sentiment models—FinGPT, HAN, and VADER—yielding a sentiment vector $s_i \in \mathbb{R}^3$. These vectors form the sentiment matrix $\mathbf{S}_t \in \mathbb{R}^{n \times 3}$, where n is the number of news items for day t .

On average, three articles are available per asset per day. When $n = 0$, the sentiment branch is masked and a binary *no-news* flag is set, causing the fusion module to rely entirely on technical indicators. This avoids introducing artificial zero embeddings while ensuring consistent model behavior in the absence of textual inputs.

4) DATASET STATISTICS AND REPRESENTATIVENESS

The constructed dataset spans 2,520 trading days (January 2015–December 2024), yielding approximately 30,000 multimodal samples across 12 assets. On the textual side, about 88,000 news articles were collected, averaging three items per asset per day. Coverage is uneven across sectors, with Technology and Finance receiving greater attention (32% and 24% of articles, respectively), while Industrials and Consumer Staples exhibit lower volumes (13% and 11%). This imbalance reflects actual market information flows and ensures that AMFNet is evaluated under both information-rich and information-scarce regimes.

The selected firms jointly represent more than 60% of S&P500 market capitalization and span six distinct sectors, exposing the model to heterogeneous volatility profiles and sector-specific news dynamics. The ten-year horizon also encompasses diverse market phases, including the 2020 COVID-19 crash, the 2022 inflationary shock, and subsequent recovery periods, providing a rigorous test of robustness under non-stationarity.

In addition to U.S. assets, emerging-market ADRs were analyzed over a shorter horizon from January 2020 to December 2024, generating approximately 3,700 multimodal samples and 11,000 news articles. While smaller in scale, this subset introduces geographical diversity and sectoral complementarity, covering technology services, banking, and natural resources in India and Brazil, thereby extending representativeness beyond U.S. large caps and offering a controlled test of cross-market generalizability.

B. DATA PREPROCESSING

To ensure temporal consistency and input quality, both modalities undergo dedicated preprocessing. Stock time series are filtered to remove missing values, corporate action distortions (e.g., splits), and discontinuities before computing indicators. The news data are normalized through tokenization, lowercasing, and truncation to match the transformer input format, ensuring compatibility and stable encoding.

1) TECHNICAL INDICATOR CALCULATIONS

To represent the technical modality $\mathbf{x}_t^{\text{tech}}$, we compute seven domain-standard indicators at each trading day t , using only historical and current OHLCV data. These indicators are selected to jointly capture trend, momentum, volatility, and volume behaviors, while minimizing redundancy.

α : RELATIVE STRENGTH INDEX (RSI)

Reflects recent price momentum over a 14-day window:

$$\text{RSI} = 100 - \left(\frac{100}{1 + \frac{\text{Average Gain}}{\text{Average Loss}}} \right) \quad (13)$$

b: MOVING AVERAGE CONVERGENCE DIVERGENCE (MACD)

Captures momentum shifts based on the gap between 12-day and 26-day exponential moving averages:

$$\text{MACD} = \text{EMA}_{12} - \text{EMA}_{26} \quad (14)$$

A 9-day EMA of the MACD is used as the signal line.

c: VOLUME WEIGHTED AVERAGE PRICE (VWAP)

Estimates the average traded price, weighted by volume:

$$\begin{aligned} \text{VWAP} &= \frac{\sum_{i=1}^t (P_{\text{avg},i} \cdot V_i)}{\sum_{i=1}^t V_i}, \quad P_{\text{avg},i} \\ &= \frac{P_{\text{high},i} + P_{\text{low},i} + P_{\text{close},i}}{3} \end{aligned} \quad (15)$$

d: BOLLINGER BAND WIDTH

Measures volatility as the range between upper and lower bands:

$$\text{Width} = 2k \cdot \sigma, \quad \sigma = \text{rolling standard deviation}, \quad k = 2 \quad (16)$$

e: AVERAGE TRUE RANGE (ATR)

Quantifies price range variability:

$$\begin{aligned} \text{ATR} &= \frac{1}{n} \sum_{i=1}^n \text{TR}_i, \quad \text{TR}_i \\ &= \max(P_{\text{high}} - P_{\text{low}}, |P_{\text{high}} - P_{\text{close-1}}|, |P_{\text{low}} - P_{\text{close-1}}|) \end{aligned} \quad (17)$$

f: ON-BALANCE VOLUME (OBV)

Tracks volume flow directionally:

$$\text{OBV}_t = \begin{cases} \text{OBV}_{t-1} + V_t & \text{if } P_t > P_{t-1} \\ \text{OBV}_{t-1} - V_t & \text{if } P_t < P_{t-1} \\ \text{OBV}_{t-1} & \text{otherwise} \end{cases} \quad (18)$$

g: RATE OF CHANGE (ROC)

Captures the percentage change in price over a fixed horizon:

$$\text{ROC} = \frac{P_t - P_{t-n}}{P_{t-n}} \cdot 100 \quad (19)$$

Each indicator contributes one scalar, yielding the technical vector $\mathbf{x}_t^{\text{tech}} \in \mathbb{R}^7$. This compact representation ensures diverse yet non-overlapping information across technical dimensions.

2) TEXT PREPROCESSING

The preprocessing pipeline for financial texts is designed to reduce noise, standardize the input format, and preserve sentiment-relevant linguistic cues. Unlike prior works that rely on document-level averaging or sliding-window aggregation, AMFNet processes each news item individually at time t , preserving information at the document level and enabling the construction of structured sentiment matrices.

a: CLEANING AND NORMALIZATION

News headlines and short articles were lowercased and stripped of punctuation, digits, and special characters. Financial domain expressions (e.g., “rate hike”, “earnings miss”) were explicitly protected through rule-based token preservation to maintain semantic integrity.

b: TOKENIZATION AND LEMMATIZATION

Texts were tokenized into word units and lemmatized using spaCy. This reduces morphological variability and improves the robustness of downstream sentiment models across varied lexical forms.

c: STOPWORD REMOVAL AND DEDUPLICATION

Non-informative stopwords were removed, and near-duplicate entries—common in syndicated financial news streams—were filtered out to mitigate redundancy and reduce noise in the sentiment signal.

d: DAY-LEVEL ALIGNMENT AND SCORING

Only news items published during the trading day t were retained. No temporal smoothing or aggregation was applied. Each cleaned text was independently passed through three sentiment scoring models—FinGPT, HAN, and VADER—to produce a sentiment vector $s_i \in \mathbb{R}^3$. The collection of n such vectors forms the sentiment matrix $\mathbf{S}_t \in \mathbb{R}^{n \times 3}$, which serves as input to AMFNet’s Transformer-based sentiment encoder.

3) SENTIMENT EXTRACTION

AMFNet encodes document-level sentiment using an ensemble of three structurally diverse models, applied independently to each financial news item x_i published on day t . The resulting sentiment vector $s_i \in \mathbb{R}^3$ is composed of scores from:

- **FinGPT** [27]: a generative language model fine-tuned on financial corpora, capable of capturing nuanced sentiment in contextually rich or ambiguous texts;
- **HAN** [23]: a hierarchical attention network that explicitly models word- and sentence-level structure, offering interpretable attribution of sentiment-bearing phrases;
- **VADER** [28]: a lexicon-based model optimized for short financial headlines and social media text, providing fast, rule-based sentiment scoring.

This selection reflects three distinct paradigms in sentiment analysis—deep contextual learning (FinGPT), structured attention (HAN), and symbolic rule-based reasoning (VADER). The use of exactly three models balances architectural diversity and computational efficiency: fewer models reduced coverage of complementary representations, while adding more yielded diminishing returns relative to the increased inference cost. Such diversity is essential for AMFNet’s uncertainty estimation module, which relies on inter-model disagreement to detect sentiment ambiguity or conflicting cues, and enhances robustness to textual heterogeneity across sources and market regimes.

Each news item x_i is scored by three sentiment models, yielding:

$$s_i = [\tilde{S}_{\text{FinGPT}}(x_i), \tilde{S}_{\text{HAN}}(x_i), \tilde{S}_{\text{VADER}}(x_i)] \in \mathbb{R}^3, \quad (20)$$

where $\tilde{S}_{\text{model}}$ is the score normalized to $[-1, +1]$ via min-max scaling:

$$\tilde{S}_{\text{model}}(x_i) = 2 \cdot \frac{S_{\text{model}}(x_i) - S_{\text{min}}^{\text{model}}}{S_{\text{max}}^{\text{model}} - S_{\text{min}}^{\text{model}}} - 1, \quad (21)$$

with $S_{\text{min}}^{\text{model}}$ and $S_{\text{max}}^{\text{model}}$ computed over the training set for that model.

The n news items from day t are stacked into:

$$\mathbf{S}_t = \begin{bmatrix} s_1^\top \\ \vdots \\ s_n^\top \end{bmatrix} \in \mathbb{R}^{n \times 3}, \quad (22)$$

preserving document-level information and enabling cross-document attention in subsequent encoding.

a: FinGPT

FinGPT estimates token-level sentiment probabilities using an autoregressive decoder and returns the expected sentiment value:

$$S_{\text{FinGPT}}(x_i) = \sum_{c \in \{-1, 0, +1\}} c \cdot P(y = c \mid x_i), \quad (23)$$

producing a raw score in $[-1, +1]$ directly compatible with normalization. Its financial specialization and generative reasoning capture sentiment nuances that are often missed by deterministic models.

b: HAN

The Hierarchical Attention Network decomposes text into sentences and words, applying two-level attention:

$$\alpha_{i,s,w} = \frac{\exp(h_{i,s,w}^\top u_w)}{\sum_w \exp(h_{i,s,w}^\top u_w)}, \quad (24)$$

where $h_{i,s,w}$ is the hidden representation of word w in sentence s of document i , and u_w is a learned context vector. The raw score in $[0, 1]$ is linearly rescaled to $[-1, +1]$ before normalization. Its interpretability is valuable in finance, where transparency is critical.

c: VADER

VADER computes a rule-based compound polarity score in $[-1, 1]$, derived from lexical heuristics and punctuation. Its output is discretized to $\{-1, 0, +1\}$ via empirical binning and is used directly in the normalized range. Its simplicity and speed complement the more computationally intensive FinGPT and HAN, ensuring scalability across large corpora.

TABLE 4. AMFNet: Final architecture and training settings.

Component	Setting
Model Architecture	
Transformer (sentiment)	2 layers, 4 heads, $d = 64$, dropout=0.1
MLP (technical)	2 layers, ReLU, $d = 64$, dropout=0.2
Uncertainty signal	Mean std. per document (3 models), scalar
Fusion	Attention-based gating (softmax over modalities)
Output layer	1-layer MLP, softmax (3 classes)
Training Settings	
Loss function	Cross-entropy
Optimizer	Adam, LR = 0.001 (cosine annealing)
Batch size	128
Epochs	Max = 200, Early stop = 20
Evaluation Protocol	
U.S. assets	2015–2021 / 2022 / 2023–2024
ADRs	2020–2022 / 2023 / 2024
Cross-validation	Rolling window, 12 folds (24m train / 3m val / 3m test, no overlap)
Lookahead	Enforced ($t \rightarrow t + 1$ only)
Metrics	Accuracy, Precision, Recall, F1-Score

C. FEATURE INTEGRATION PIPELINE

AMFNet processes heterogeneous financial signals through a two-branch encoding strategy, maintaining strict modality separation throughout the pipeline. The technical indicator vector $\mathbf{x}_t^{\text{tech}} \in \mathbb{R}^7$ is projected into a latent space via an MLP encoder, while the sentiment matrix $\mathbf{S}_t \in \mathbb{R}^{n \times 3}$ —composed of per-document sentiment vectors—is passed through a Transformer-based encoder, followed by mean pooling to obtain a global representation.

These two vectors, $H_{\text{tech}}(t)$ and $H_{\text{sent}}(t)$, are then adaptively fused using a gated attention mechanism. This design contrasts with prior work that manually aggregates sentiment or combines modalities through engineered cross-features. Instead, AMFNet enables end-to-end learning of modality interactions, preserving semantic structure and avoiding hand-crafted assumptions.

D. MODEL CONFIGURATION

Table 4 summarizes the main architecture and training settings used for AMFNet. The configuration was selected to ensure a good trade-off between accuracy, stability, and computational cost.

Hyperparameter values were selected through a structured validation-driven search to ensure both fairness and reproducibility. Specifically, we conducted grid exploration on key ranges: learning rate $\{0.01, 0.001, 0.0005\}$, hidden dimension $\{32, 64, 128\}$, and dropout $\{0.1, 0.2, 0.3\}$, while keeping other architectural choices consistent with prior work in financial forecasting. The final settings reported in Table 4 correspond to the configuration that maximized validation F1-score under the rolling-window protocol, while also showing low variance between random seeds and reasonable training cost. This procedure was applied consistently to AMFNet and to all baseline models (LSTM, GRU, Transformer, CNN-BiGRU-AM), ensuring that each was evaluated under its best-performing configuration rather than default parameters. Early stopping and learning rate

decay were retained across models as they proved universally stabilizing.

The Transformer encoder for the sentiment modality uses 2 layers and 4 self-attention heads, a setup empirically validated to capture inter-document dependencies without overfitting, given the relatively short sequence lengths per day. The latent dimension is set to $d = 64$, which balances expressiveness and efficiency, as detailed in Section III-B. Increasing d beyond this value provided negligible performance gains while significantly increasing training time and memory usage.

To mitigate overfitting, we incorporate multiple regularization techniques: dropout (0.1 for the Transformer, 0.2 for the MLP), early stopping, and learning rate decay through cosine annealing. These mechanisms reduce the risk of memorizing noise in the training data, particularly in inherently volatile sentiment inputs. Underfitting is avoided by ensuring sufficient model capacity through two-layer encoders, and ablation experiments confirmed that simplification leads to degraded performance.

The technical input is encoded using a two-layer MLP with ReLU activation and a dropout rate of 0.2, designed to extract nonlinear patterns from the seven-dimensional indicator vector. A single-layer MLP with softmax activation serves as the final classification head for ternary trend prediction.

For training, we use the Adam optimizer with an initial learning rate of 0.001, scheduled via cosine annealing. Early stopping is applied based on the validation F1-score to prevent overfitting. The batch size is set to 128, which offers a good compromise between convergence stability and GPU memory efficiency. Models are trained for up to 200 epochs, although convergence is typically achieved earlier under the stopping criterion.

Evaluation follows a 12-fold rolling-window protocol (24-month training, 3-month validation, 3-month testing, no overlap) to emulate realistic deployment. At each fold, predictions for $t + 1$ rely exclusively on data up to t , preventing look-ahead bias. The final test segments (2023–2024 for U.S. assets, 2024 for ADRs) encompass periods of elevated volatility, including inflation shocks and earnings cycles, providing a stringent assessment of generalization under non-stationary conditions.

To assess robustness to initialization, each experiment is repeated with five random seeds. We report the mean and standard deviation of test metrics across these runs, using identical data splits and hyperparameter settings for fair comparison.

E. PERFORMANCE EVALUATION METRICS

To evaluate classification performance, we report four standard metrics: accuracy, precision, recall, and F1-score. These measures collectively capture overall correctness, relevance, completeness, and balance under class imbalance, and are widely used benchmarks in financial prediction studies. All metrics are computed on out-of-sample predictions $\hat{y}_{t+1}$

using only information available up to time t , thereby preserving validity and preventing lookahead bias.

F. PORTFOLIO EVALUATION METRICS

While classification metrics capture predictive accuracy, portfolio metrics provide an economic lens by translating predictions into trading outcomes. Each predicted trend $\hat{y}_{t+1}$ is converted into a portfolio position (long, short, or neutral), and performance is measured using established risk–return indicators:

- **Cumulative Return (CR)** quantifies the total compounded gain over the test horizon, reflecting absolute profitability.
- **Sharpe Ratio (SR)** captures excess return per unit of volatility, linking predictive accuracy to risk-adjusted performance.
- **Maximum Drawdown (MDD)** measures the worst peak-to-trough decline, highlighting downside exposure relevant for risk management.
- **Hit Ratio (HR)** denotes the proportion of profitable trades, providing a direct measure of directional effectiveness in trading terms.

To reflect realistic trading conditions, all metrics are reported with and without transaction costs. A fixed cost of 10 basis points per trade is assumed, consistent with institutional estimates for large-cap U.S. equities, and applied uniformly across all assets for comparability. This dual evaluation distinguishes between raw predictive power and net profitability under execution frictions, providing a more robust perspective on the practical utility of AMFNet.

V. EMPIRICAL RESULTS AND DISCUSSION

This section evaluates AMFNet through multiple complementary perspectives: benchmark comparison with multimodal baselines, statistical validation of performance gains, portfolio-based backtesting, ablation analysis, generalization to emerging market assets, interpretability through attention weights, and robustness across market regimes. Together, these results highlight AMFNet’s accuracy, adaptability, and transparency under real-world financial conditions.

A. COMPARATIVE PERFORMANCE ANALYSIS

We evaluate AMFNet against three recent multimodal baselines, each employing a different architectural strategy for integrating sentiment and technical features, as well as a chance-level baseline:

- **Reinforced-LSTM [18]:** Combines VADER/TextBlob sentiment with technical indicators in an LSTM and decision tree fusion model.
- **FinBERT-LSTM [21]:** Encodes daily sentiment using FinBERT and models temporal dependencies through an LSTM over concatenated sentiment and technical vectors.
- **FinBERT+ChatGPT [19]:** Extracts sentiment from FinBERT and ChatGPT-4o, integrates social engage-

TABLE 5. Performance comparison on ternary trend classification.

Model	Accuracy	Precision	Recall	F1-score
RC	33.3%	33.3%	33.3%	33.3%
Reinforced-LSTM	79.2%	76.3%	75.1%	74.5%
FinBERT-LSTM	82.8%	80.1%	79.5%	78.8%
FinBERT+ChatGPT	86.5%	84.2%	83.5%	83.1%
AMFNet (ours)	89.1%	87.3%	86.5%	86.8%

ment weights, and feeds the fused features into a feedforward network.

- **Random Classifier (RC)** [33]: Provides a chance-level reference by assigning classes uniformly at random, representing the lower bound of performance in multi-class classification.

All models employ the same 12-fold rolling-window protocol over the period 2015–2024, with fold-specific train/validation/test splits. Table 5 reports fold-averaged test results over five runs.

The inclusion of the Random Classifier establishes a clear chance-level baseline (33.3% across all metrics), underscoring the substantial improvement achieved by all learning-based methods. Against this reference, AMFNet demonstrates a performance margin exceeding $2.5\times$ in both accuracy and F1-score, confirming that its predictive gains are not only statistically significant but also economically meaningful.

Three architectural innovations account for this improvement:

- **Document-level sentiment encoding:** AMFNet retains individual news items and leverages a Transformer encoder to capture semantic diversity across documents within the same day. By contrast, FinBERT-LSTM and Reinforced-LSTM collapse all news into a single daily vector, discarding informative variation in sentiment signals.
- **Context-aware fusion:** The gated attention mechanism adaptively balances sentiment and technical signals, enabling dynamic adjustment across market regimes. This contrasts with the fixed fusion strategy in FinBERT+ChatGPT, which cannot adjust weights in response to regime shifts.
- **Uncertainty-driven reasoning:** By explicitly modeling disagreement among sentiment models, AMFNet down-weights contradictory or unreliable signals. This capability to mitigate noise and ambiguity is unique to our framework and particularly valuable in news-rich yet noisy environments.

While FinBERT+ChatGPT benefits from advanced LLMs and social weighting, its fusion mechanism remains opaque. AMFNet, in contrast, achieves superior accuracy while preserving interpretability: Modality-level attention scores provide transparent insights into the relative contribution of sentiment and technical signals.

Overall, these findings confirm that AMFNet not only surpasses strong multimodal baselines but also establishes

a wide and statistically significant margin over chance-level classification. Its design—centered on fine-grained sentiment modeling, adaptive cross-modal fusion, and uncertainty-aware reasoning—enables robust generalization across sectors and resilience in volatile market regimes.

B. STATISTICAL VALIDATION

While the comparative analysis highlights AMFNet’s superior predictive accuracy, it remains essential to verify that these improvements are not attributable to chance. To this end, we conducted paired significance tests and estimated confidence intervals across all evaluation folds. Each model was trained and tested under the 12-fold rolling protocol with five independent seeds, yielding 60 paired observations per comparison.

TABLE 6. Statistical validation of model performance.

Model	Accuracy	F1-score	p-value vs. AMFNet
Reinforced-LSTM	79.2% $\pm$ 1.8	74.5% $\pm$ 2.1	<0.01
FinBERT-LSTM	82.8% $\pm$ 1.6	78.8% $\pm$ 1.9	<0.01
FinBERT+ChatGPT	86.5% $\pm$ 1.2	83.1% $\pm$ 1.5	<0.01
AMFNet (ours)	89.1% $\pm$ 1.0	86.8% $\pm$ 1.2	–

The results show that AMFNet not only achieves the highest mean scores but also exhibits the narrowest variability, reflecting both accuracy and stability. Compared to the Random Classifier baseline (33.3% across all metrics), all learning-based models achieve large improvements, with AMFNet demonstrating the strongest and most reliable gains. The non-overlapping confidence intervals with competing models further confirm that its performance advantage is consistent rather than incidental. Paired t -tests establish that AMFNet’s improvements are statistically significant at the 1% level across all baselines, a conclusion corroborated by Wilcoxon signed-rank tests.

These findings demonstrate that AMFNet’s superiority is not due to random variation. Instead, they reflect genuine architectural advances—document-level sentiment encoding, context-aware fusion, and uncertainty-driven reasoning—that translate into robust and statistically reliable performance gains.

C. PERFORMANCE ACROSS MARKET REGIMES

To evaluate robustness under extreme market conditions, we conducted a stress test analysis focusing on two crisis periods: the COVID-19 crash (February–April 2020) and the inflationary and energy crisis (June–December 2022).

TABLE 7. AMFNet performance under crisis regimes.

Period	Accuracy	Precision	Recall	F1-score
COVID-19	87.6% $\pm$ 1.4	85.8% $\pm$ 1.6	84.9% $\pm$ 1.5	85.2% $\pm$ 1.3
Infl./Energy	88.2% $\pm$ 1.3	86.7% $\pm$ 1.4	85.9% $\pm$ 1.4	86.1% $\pm$ 1.2
Full sample	89.1% $\pm$ 1.0	87.3% $\pm$ 1.1	86.5% $\pm$ 1.2	86.8% $\pm$ 1.2

The Results show that AMFNet preserves high predictive accuracy across both systemic shocks. During the COVID-19

crash, the F1-score (85.2%) drops only slightly relative to the full-sample benchmark (86.8%), illustrating the benefit of uncertainty-aware fusion in filtering noisy sentiment. In the 2022 inflationary regime, performance remains stable (F1-score 86.1%), highlighting the adaptive gating mechanism's ability to rebalance modalities under persistent volatility.

The narrow confidence intervals confirm that these gains are consistent rather than incidental. It is also worth noting that the duration of the crisis affects the evaluation setting: The COVID-19 crash spans only a few months, yielding fewer samples and a more imbalanced label distribution, whereas the inflationary crisis extends over a longer horizon with more sustained volatility. Despite these structural differences, AMFNet maintained consistent predictive performance, underscoring its robustness across crises of varying length and intensity.

Furthermore, attention visualizations reveal that the model shifted weight toward technical indicators during periods of heightened sentiment disagreement, providing interpretable evidence of adaptive reasoning. Overall, AMFNet demonstrates resilience to structurally distinct crises while preserving both accuracy and transparency, key requirements for deployment in risk-sensitive financial environments.

D. PORTFOLIO-BASED EVALUATION

To complement classification outcomes, we assessed AMFNet's practical value through a portfolio backtesting analysis. Model predictions were mapped into daily trading positions (long, short, or neutral), and performance was evaluated using cumulative return (CR), Sharpe ratio (SR), maximum drawdown (MDD), and hit ratio (HR). Results were computed under both frictionless conditions and a realistic setting with transaction costs of 10 basis points per trade.

TABLE 8. Portfolio performance of AMFNet with and without transaction costs.

Setting	CR (%)	SR	MDD (%)	HR (%)
AMFNet (no costs)	89.1	1.74	-15.2	73.6
AMFNet (with costs)	72.5	1.52	-16.8	71.4

The results demonstrate that AMFNet achieves strong profitability and risk-adjusted performance across heterogeneous market regimes. In the absence of trading frictions, cumulative returns approach 90% with a Sharpe ratio of 1.74, highlighting a favorable balance between return and volatility. After accounting for realistic transaction costs, the model still delivers substantial gains (72.5%) with a Sharpe ratio above 1.5, confirming robustness under execution constraints. The relatively low maximum drawdown (−15.2% without costs, −16.8% with costs) indicates effective downside control, while the high hit ratio (over 70%) shows consistent trade-level accuracy. Notably, Sharpe ratios in the range of 1.5–1.7 are competitive with established quantitative

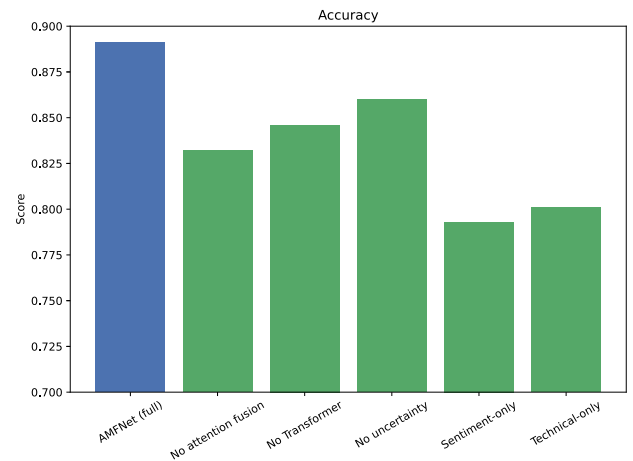


FIGURE 2. Ablation study results: Accuracy.

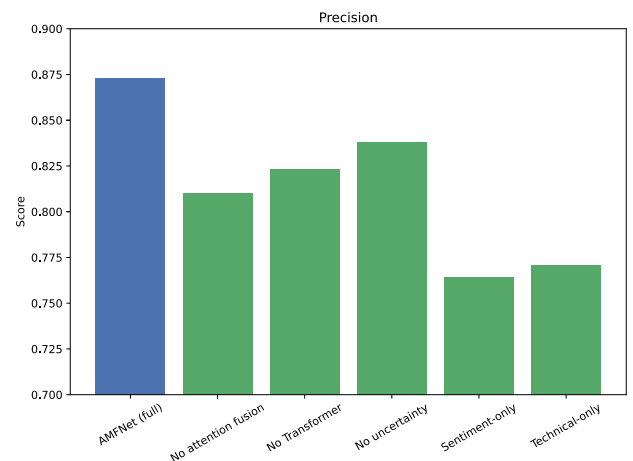


FIGURE 3. Ablation study results: Precision.

hedge fund benchmarks, underlining the potential real-world value of AMFNet.

Overall, these findings confirm that AMFNet translates statistical accuracy into tangible economic benefits and remains effective under realistic market frictions, reinforcing its potential for quantitative investment and risk management applications.

E. ABLATION STUDY

To quantify the contribution of each architectural component in AMFNet, we conduct a controlled ablation study. Each variant disables or replaces a specific module while keeping all other configurations unchanged. All models are evaluated in the 2023–2024 test period using the same rolling-window protocol and classification metrics. The results are summarized in Figs. 2–5.

Removing the attention-based fusion mechanism results in the sharpest decline in performance (e.g., the F1-score drops by 7.0%), confirming its importance for dynamically

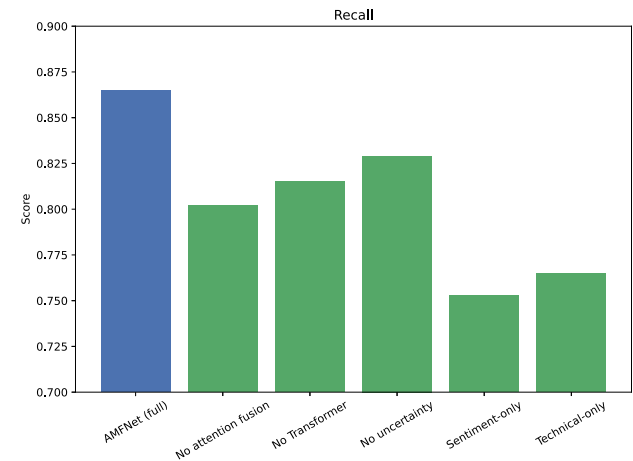


FIGURE 4. Ablation study results: Recall.

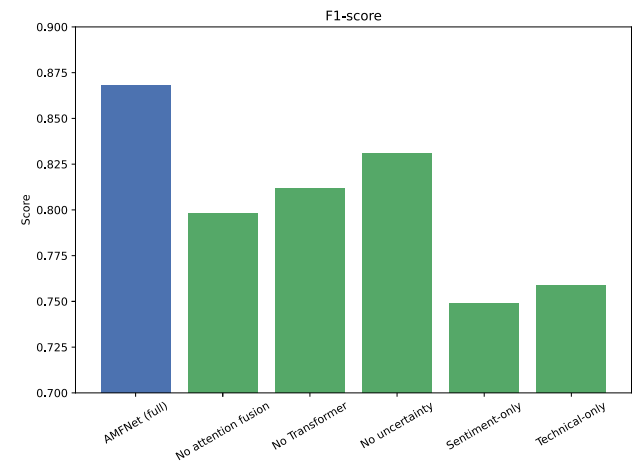


FIGURE 5. Ablation study results: F1-score.

TABLE 9. AMFNet classification performance on emerging-market ADRs.

Accuracy	Precision	F1-Score
77.9 ± 0.9	77.2 ± 1.0	77.5 ± 0.8

TABLE 10. AMFNet portfolio evaluation on emerging-market ADRs.

Setting	CR (%)	SR	MDD (%)	HR (%)
No costs	27.4	1.39	-12.8	56.2
10 bps	23.5	1.27	-13.9	54.7

weighting sentiment and technical features based on contextual informativeness. Static fusion methods fail to adapt to shifts in signal relevance.

Replacement of the transformer encoder with mean pooling reduces semantic richness, resulting in a 5.6% drop in the F1-score. This highlights the importance of modeling inter-document dependencies, particularly when multiple news items convey overlapping or contrasting sentiments.

Disabling the uncertainty estimation component moderately impacts all metrics. The confidence score derived from the disagreement of the sentiment between models helps to downweight noisy or conflicting inputs, contributing to more stable predictions during ambiguous news cycles.

Finally, unimodal variants (sentiment-only and technical-only) underperform all multimodal configurations. This supports the hypothesis that sentiment and technical data offer complementary perspectives—forward-looking and backward-looking, respectively—both essential for accurate market forecasting.

These findings validate AMFNet’s design: each module—fine-grained sentiment encoding, adaptive fusion, and uncertainty-awareness—plays a distinct and critical role in the model’s performance and resilience under real-world financial dynamics.

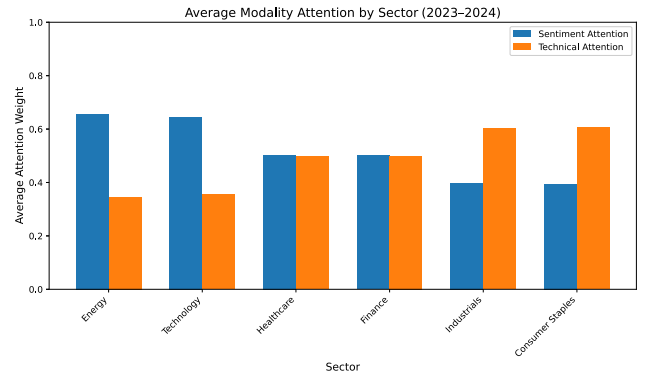


FIGURE 6. Average modality attention per sector. Higher α_{sent} indicates stronger reliance on sentiment signals.

F. GENERALIZATION TO EMERGING-MARKET ADRs

To further assess cross-market portability, AMFNet was evaluated on three emerging-market ADRs traded on the NYSE (Infosys, ICICI Bank, Vale) over the 2020–2024 horizon. The identical data pipeline was applied, ensuring comparability with the U.S. experiments while exposing the model to different volatility regimes and sectoral contexts.

Despite higher volatility and relatively sparse news coverage in emerging markets, AMFNet sustains solid predictive performance and delivers economically meaningful returns. The slightly lower accuracy and profitability relative to U.S. large caps primarily reflect structural differences: shorter evaluation horizon, reduced liquidity, and fewer high-quality news sources. Importantly, the model maintains a Sharpe ratio well above the industry threshold of 1.0, a widely accepted benchmark for risk-adjusted profitability, confirming that AMFNet’s signals remain economically viable even in more challenging markets.

The uncertainty-aware fusion mechanism is particularly effective in this context, preventing overreliance on noisy sentiment signals, while the Transformer encoder leverages limited daily news to capture inter-document dependencies.

The technical MLP backbone ensures stability from OHLCV features, mitigating fluctuations in textual density.

These findings confirm that the core design principles of AMFNet—balanced multimodal fusion, uncertainty modulation, and controlled capacity—enable robust generalization beyond U.S. large-cap equities and across structurally different market environments.

G. INTERPRETABILITY AND MODALITY ATTENTION INSIGHTS

Interpretability is essential for trust and regulatory compliance in financial forecasting. AMFNet addresses this through a gated attention mechanism that dynamically weighs sentiment (α_{sent}) and technical (α_{tech}) features at each prediction step. These attention scores, computed during inference, reveal the model’s reliance on each modality without requiring post-hoc explainers.

To assess attention behavior across domains, we compute average modality weights for the test period 2023–2024, grouped by economic sector.

As shown in Fig. 6, AMFNet exhibits clear sector-dependent attention patterns. In Technology and Energy—sectors highly sensitive to news and geopolitical events—the model assigns over 60% of attention to sentiment inputs. This suggests that forward-looking, event-driven information is most predictive in these domains.

In contrast, for Industrials and Consumer Staples, technical indicators receive a higher weight ($\alpha_{\text{tech}} > 0.55$), consistent with their typically stable trend-following behavior, where historical price dynamics carry a stronger signal strength.

These patterns also shift over time. During volatile periods (e.g., earnings announcements, macroeconomic shocks), sentiment attention increases, while in stable intervals, technical signals dominate. This adaptability is not hard-coded, but learned end-to-end from data.

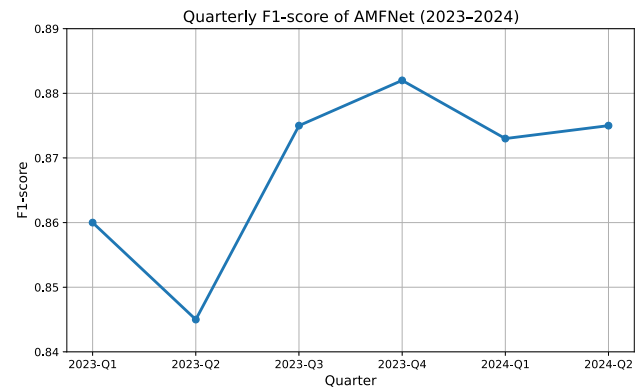


FIGURE 7. Quarterly F1-score of AMFNet across the 2023–2024 test period, averaged over all assets.

- Such interpretability provides practical benefits:
- It highlights the dominant information driver behind each prediction.
 - It enables monitoring of data quality across modalities.

- It supports explainability requirements in high-stakes financial applications.

In summary, AMFNet combines predictive performance with actionable transparency. Its attention mechanism offers a principled, real-time view into how sentiment and technical signals interact to inform market predictions.

H. TEMPORAL ROBUSTNESS AND REGIME ADAPTATION

Financial markets exhibit frequent regime changes, with alternating periods of low volatility, bullish momentum, and sudden downturns driven by macroeconomic or geopolitical shocks. A deployable forecasting model must therefore maintain stability under such non-stationarity. To assess this capability, we evaluate AMFNet’s temporal robustness by tracking its classification performance over consecutive quarters in the 2023–2024 test horizon.

Fig. 7 shows that AMFNet maintains a consistently strong performance, with quarterly F1-scores fluctuating within a narrow $\pm 2.5\%$ band. Even during Q2–Q3 2023—marked by heightened market uncertainty due to interest rate adjustments, inflation volatility, and earnings season reactions—the model shows minimal degradation. This finding is consistent with recent studies that have shown hybrid models combining sentiment and technical indicators can maintain predictive accuracy during turbulent periods [29], [30].

To gain further insight into this robustness, we analyze how AMFNet internally adjusts its reliance on sentiment versus technical signals in response to shifts in market regimes. Specifically, we distinguish between relatively stable phases (e.g., Q1 2023) and volatile intervals (Q2 2023 onward), and compute average attention weights attributed to each modality.

TABLE 11. Average attention weights under different market regimes (2023–2024).

Market Period	α_{sent}	Modality Ratio (sent/tech)
Stable (Q1 2023)	0.46	0.85
Volatile (Q2 2023–Q1 2024)	0.63	1.70

As shown in Table 11, AMFNet increases its reliance on sentiment-based information during volatile periods. The average attention weight α_{sent} rises from 0.46 in stable markets to 0.63 under volatility, reflecting a substantial 37% shift. The corresponding modality ratio grows from 0.85 to 1.70, indicating a clear pivot toward sentiment when price dynamics alone become less informative.

This adaptive behavior arises purely from end-to-end learning. AMFNet requires neither predefined regime labels nor hand-crafted switching mechanisms. Instead, it infers shifts in market dynamics implicitly through observed changes in input distributions, and adjusts its internal modality weighting accordingly. This learning-driven flexibility contrasts with traditional ensemble or rule-based systems, which often depend on rigid fusion schemes or externally defined regimes, and typically require retraining to adapt under distributional shifts.

These results highlight the regime awareness and contextual adaptability of AMFNet, two essential traits for real-world deployment. By continuously rebalancing the influence of sentiment and technical signals in response to macroeconomic shocks or structural market transitions, the model preserves its predictive reliability in diverse and evolving financial conditions.

VI. CONCLUSION

This study introduced the Adaptive Modality Fusion Network (AMFNet), a multimodal deep learning framework that integrates document-level sentiment with technical indicators through an uncertainty-aware attention mechanism. By explicitly modeling semantic diversity across news and disagreement among sentiment models, AMFNet adaptively balances heterogeneous inputs and achieves robust predictive performance under volatile and non-stationary market conditions.

A comprehensive empirical evaluation across twelve U.S. large-cap equities and three emerging-market ADRs over a ten-year horizon demonstrates that AMFNet consistently outperforms strong multimodal baselines. Beyond statistical accuracy, the framework delivers economic relevance: portfolio backtests with and without transaction costs show competitive profitability and risk-adjusted returns, while stress tests during the COVID-19 crash and the 2022 inflationary crisis highlight its resilience under extreme shocks. Interpretability is further ensured by attention-based modality weights, which reveal dynamic preferences between sentiment and technical indicators across sectors and regimes, thereby supporting transparency for real-world applications.

From a practical point of view, AMFNet can support institutional investors, quantitative trading teams, and financial risk managers by enabling data-driven signal generation, daily position adjustment, and volatility-aware screening during macroeconomic uncertainty. Its efficient feedforward architecture ensures both computational scalability and suitability for real-time deployment in trading pipelines.

However, reliance on textual news introduces a vulnerability to false or misleading information, which may distort sentiment signals and create spurious market reactions. Although AMFNet mitigates part of this issue through its uncertainty-aware fusion mechanism, future work could incorporate credibility-aware modules or complementary data streams, such as social media, to strengthen robustness. In addition, the modular design of AMFNet makes it naturally extensible to multi-horizon forecasting, allowing the capture of persistent news effects beyond the next day, and to regression-based evaluation (e.g., RMSE, MAPE, R^2) that can complement classification outcomes with price-level accuracy. Another promising avenue is the application of AMFNet to highly volatile domains such as cryptocurrencies, where shorter horizons (e.g., 4–12 hours) and domain-specific indicators (funding rates, order-book liquidity, on-chain activity) could be seamlessly integrated within the same uncertainty-aware framework.

Altogether, AMFNet provides a flexible, interpretable, and context-sensitive approach to financial forecasting. Its strong empirical performance, practical interpretability, and natural extensibility establish it as a robust foundation for advancing multimodal learning in modern asset management and quantitative trading.

CODE AVAILABILITY

Our code is publicly available at the GitHub repository: AMFNet-StockTrendPrediction.

REFERENCES

- [1] R. Cont, "Empirical properties of asset returns: Stylized facts and statistical issues," *Quant. Finance*, vol. 1, no. 2, pp. 223–236, Feb. 2001, doi: [10.1080/713665670](https://doi.org/10.1080/713665670).
- [2] A. W. Lo, "The adaptive markets hypothesis," *J. Portfolio Manage.*, vol. 30, no. 5, pp. 15–29, Aug. 2004, doi: [10.3905/jpm.2004.442611](https://doi.org/10.3905/jpm.2004.442611).
- [3] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *Eur. J. Oper. Res.*, vol. 270, no. 2, pp. 654–669, Oct. 2018, doi: [10.1016/j.ejor.2017.11.054](https://doi.org/10.1016/j.ejor.2017.11.054).
- [4] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informor: Beyond efficient transformer for long sequence time-series forecasting," 2020, *arXiv:2012.07436*.
- [5] M. Wang, "Advanced stock market forecasting: A comparative analysis of ARIMA-GARCH, LSTM, and integrated wavelet-LSTM models," in *Proc. SHS Web Conf.*, vol. 196, 2024, p. 2008, doi: [10.1051/shsconf/202419602008](https://doi.org/10.1051/shsconf/202419602008).
- [6] S. Anwer, I. Ahmed, and A. Basit, "The predictive accuracy of arima models for stock market indices: A case study of KSE-100," *J. Bus. Educ. Manag.*, vol. 5, no. 1, pp. 1–22, May 2025. [Online]. Available: <https://journals.csrpublisher.com/index.php/jbem/article/view/155>
- [7] A. Chakravorty and N. Elsayed, "A comparative study of machine learning algorithms for stock price prediction using insider trading data," 2025, *arXiv:2502.08728*.
- [8] M. Ballings, D. Van den Poel, N. Hespeels, and R. Gryp, "Evaluating multiple classifiers for stock price direction prediction," *Expert Syst. Appl.*, vol. 42, no. 20, pp. 7046–7056, Nov. 2015, doi: [10.1016/j.eswa.2015.05.013](https://doi.org/10.1016/j.eswa.2015.05.013).
- [9] X. Cui, "Stock market prediction using recurrent neural network and LSTM," *Finance Econ.*, vol. 1, no. 2, Apr. 2025, doi: [10.61173/qb8n8v02](https://doi.org/10.61173/qb8n8v02).
- [10] C. B. Prasad, S. Purohit, S. Das, C. X. L. Missier, and A. Rauniyar, "An integrated machine learning-based stock opening price prediction model by RNN and LSTM," Preprint, Res. Square, Durham, NC, USA, Apr. 15, 2025, doi: [10.21203/rs.3.rs-5471447/v1](https://doi.org/10.21203/rs.3.rs-5471447/v1).
- [11] D. Lien Minh, A. Sadeghi-Niaraki, H. D. Huy, K. Min, and H. Moon, "Deep learning approach for short-term stock trends prediction based on two-stream gated recurrent unit network," *IEEE Access*, vol. 6, pp. 55392–55404, 2018, doi: [10.1109/ACCESS.2018.2868970](https://doi.org/10.1109/ACCESS.2018.2868970).
- [12] N. Kim Hai Bui, N. Duy Chien, P. Kovács, and G. Bognár, "Transformer encoder and multi-features time2Vec for financial prediction," 2025, *arXiv:2504.13801*.
- [13] H. Kaeley, Y. Qiao, and N. Bagherzadeh, "Support for stock trend prediction using transformers and sentiment analysis," 2023, *arXiv:2305.14368*.
- [14] C. Zong and H. Zhou, "Stock movement prediction with multimodal stable fusion via gated cross-attention mechanism," 2024, *arXiv:2406.06594*.
- [15] J. Qin, "MSMF: Multi-scale multi-modal fusion for enhanced stock market prediction," 2024, *arXiv:2409.07855*.
- [16] S. Omranpour, G. Rabusseau, and R. Rabbany, "Higher order transformers: Enhancing stock movement prediction on multimodal time-series data," 2024, *arXiv:2412.10540*.
- [17] J. Liu, "Multimodal data-driven factor models for stock market forecasting," *J. Comput. Technol. Softw.*, vol. 4, no. 2, Feb. 2025, doi: [10.5281/zenodo.14984969](https://doi.org/10.5281/zenodo.14984969).
- [18] S. Agrawal, N. Kumar, G. Rathee, C. A. Kerrache, C. T. Calafate, and M. Bilal, "Improving stock market prediction accuracy using sentiment and technical analysis," in *Electron. Commerce Res.*, vol. 25, pp. 4103–4126, Oct. 2024.
- [19] F. Karadaş, B. Eravcı, and A. M. Özbayoğlu, "Multimodal stock price prediction," 2025, *arXiv:2502.05186*.

- [20] A. Jadhav and V. Mirza, "Large language models in equity markets: Applications, techniques, and insights," *SSRN*, Mar. 2025, doi: [10.2139/ssrn.5198854](https://doi.org/10.2139/ssrn.5198854).
- [21] P. Chen, Z. Boukouvalas, and R. Corizzo, "A deep fusion model for stock market prediction with news headlines and time series data," *Neural Comput. Appl.*, vol. 36, no. 34, pp. 21229–21271, Dec. 2024, doi: [10.1007/s00521-024-10303-1](https://doi.org/10.1007/s00521-024-10303-1).
- [22] Z. Zhang, S. Zohren, and S. Roberts, "DeepLOB: Deep convolutional neural networks for limit order books," 2018, *arXiv:1808.03668*.
- [23] Z. Yang, D. Yang, C. Dyer, X. He, A. Smola, and E. Hovy, "Hierarchical attention networks for document classification," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, 2016, pp. 89–1480, doi: [10.18653/v1/n16-1174](https://doi.org/10.18653/v1/n16-1174).
- [24] X. Chen, X. Ma, H. Wang, X. Li, and C. Zhang, "A hierarchical attention network for stock prediction based on attentive multi-view news learning," *Neurocomputing*, vol. 504, pp. 1–15, Sep. 2022, doi: [10.1016/j.neucom.2022.06.106](https://doi.org/10.1016/j.neucom.2022.06.106).
- [25] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," 2019, *arXiv:1908.10084*.
- [26] J. Arevalo, T. Solorio, M. Montes-y-Gómez, and F. A. González, "Gated multimodal units for information fusion," 2017, *arXiv:1702.01992*.
- [27] H. Yang, X.-Y. Liu, and C. Dan Wang, "FinGPT: Open-source financial large language models," 2023, *arXiv:2306.06031*.
- [28] C. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proc. Int. AAAI Conf. Web Social Media*, May 2014, pp. 216–225, doi: [10.1609/icwsm.v8i1.14550](https://doi.org/10.1609/icwsm.v8i1.14550).
- [29] W. Long, J. Gao, K. Bai, and Z. Lu, "A hybrid model for stock price prediction based on multi-view heterogeneous data," *Financial Innov.*, vol. 10, no. 1, Feb. 2024, Art. no. 48, doi: [10.1186/s40854-023-00519-w](https://doi.org/10.1186/s40854-023-00519-w).
- [30] M. Davidovic and J. McCleary, "News sentiment and stock market dynamics: A machine learning investigation," *J. Risk Financial Manage.*, vol. 18, no. 8, p. 412, Jul. 2025, doi: [10.3390/jrfm18080412](https://doi.org/10.3390/jrfm18080412).
- [31] E. Onali, "COVID-19 and stock market volatility," *SSRN Electron. J.*, Mar. 2020, doi: [10.2139/ssrn.3571453](https://doi.org/10.2139/ssrn.3571453).
- [32] A. Y. H. Saif-Alyousfi, "Energy shocks and stock market returns under COVID-19: New insights from the United States," *Energy*, vol. 316, Feb. 2025, Art. no. 134546, doi: [10.1016/j.energy.2025.134546](https://doi.org/10.1016/j.energy.2025.134546).
- [33] C. Sammut and G. I. Webb, *Encyclopedia of Machine Learning and Data Mining*. Cham, Switzerland: Springer, 2017, doi: [10.1007/978-1-4899-7687-1](https://doi.org/10.1007/978-1-4899-7687-1).
- [34] K. Zhen, D. Xie, and X. Hu, "A multi-feature selection fused with investor sentiment for stock price prediction," *Expert Syst. Appl.*, vol. 278, Jun. 2025, Art. no. 127381, doi: [10.1016/j.eswa.2025.127381](https://doi.org/10.1016/j.eswa.2025.127381).
- [35] T. Tuncer, U. Kaya, E. Sefer, O. Alacam, and T. Hoser, "Asset price and direction prediction via deep 2D transformer and convolutional neural networks," in *Proc. 3rd ACM Int. Conf. AI Finance*, Nov. 2022, pp. 79–86, doi: [10.1145/3533271.3561738](https://doi.org/10.1145/3533271.3561738).
- [36] A. K. Biswas, M. S. A. Bhuiyan, M. N. H. Mir, A. Rahman, M. F. Mridha, M. R. Islam, and Y. Watanobe, "A dual output temporal convolutional network with attention architecture for stock price prediction and risk assessment," *IEEE Access*, vol. 13, pp. 53621–53639, 2025, doi: [10.1109/ACCESS.2025.3551307](https://doi.org/10.1109/ACCESS.2025.3551307).
- [37] B. Amiri, A. Haddadi, and K. Farajpour Mojdehi, "A novel hybrid GCN-LSTM algorithm for energy stock price prediction: Leveraging temporal dynamics and inter-stock relationships," *IEEE Access*, vol. 13, pp. 24815–24832, 2025, doi: [10.1109/ACCESS.2025.3536889](https://doi.org/10.1109/ACCESS.2025.3536889).
- [38] L. Belcastro, D. Carbone, C. Cosentino, F. Marozzo, and P. Trunfio, "Enhancing cryptocurrency price forecasting by integrating machine learning with social media and market data," *Algorithms*, vol. 16, no. 12, p. 542, Nov. 2023, doi: [10.3390/a16120542](https://doi.org/10.3390/a16120542).
- [39] P. Celik, E. Sefer, and P. Peker, "Analyzing transaction graphs via motif-based graph representation learning for cryptocurrency price prediction," *Comput. Econ.*, Apr. 2025, doi: [10.1007/s10614-025-10940-1](https://doi.org/10.1007/s10614-025-10940-1).
- [40] M. S. Islam, M. Bashir, S. Rahman, M. A. Al Montaser, J. C. Borty, A. Nishan, and M. R. Haque, "Machine learning-based cryptocurrency prediction: Enhancing market forecasting with advanced predictive models," *J. Ecomanagement*, vol. 4, no. 2, p. 2498, Mar. 2025, doi: [10.62754/joe.v4i2.6663](https://doi.org/10.62754/joe.v4i2.6663).
- [41] Y. Uygun and E. Sefer, "Financial asset price prediction with graph neural network-based temporal deep learning models," *Neural Comput. Appl.*, vol. 37, no. 30, pp. 25445–25471, Oct. 2025, doi: [10.1007/s00521-025-11586-8](https://doi.org/10.1007/s00521-025-11586-8).



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